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Real-time detection of potato spoilage by gas sensor array coupled with microbiological analysis

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ABSTRACT

Vegetable spoilage is a major contributor to global food loss, with potatoes being especially vulnerable due to their high nutrient content and microbial susceptibility. This study introduces a real-time, non-invasive method for detecting potato spoilage using a metal oxide semiconductor (MOS) gas sensor array, focusing on the MQ-3 sensor. An Arduino-based system continuously monitored gas emissions under ambient, high-humidity, and UV-treated storage conditions. Sensor data were validated against microbiological analysis, including *E. coli*, *coliforms*, total aerobic counts, yeast, and mold. A strong correlation was observed between sensor resistance decline and microbial growth, with MQ-3 showing high sensitivity to volatile organic compounds (VOCs). UV pretreatment delayed microbial activity and stabilized sensor signals. Comparative analysis revealed that only MQ-3 and MQ-135 were effective for early potato spoilage detection. These findings demonstrate the potential of low-cost MOS sensors as practical alternatives to traditional methods, enhancing food safety and reducing post-harvest losses.

1. Introduction

Globally, an estimated 14 % of the world's food production is lost between the post-harvests excluding losses at the retail stage, with vegetables being particularly vulnerable due to their perishable nature (The State of Food and Agriculture 2019, 2019). Vegetables are particularly susceptible to spoilage due to their perishable nature (ElManawy et al., 2025; Emadi & Rahmanian, 2020). According to the Food and Agriculture Organization (FAO), fruit and vegetable losses are especially high in low- and middle-income countries. This is due to factors such as inadequate cold storage and poor transportation infrastructure, which lead to infections and physical damage to the produce. This results in millions of tons of spoiled produce globally each year (Kanjilal et al., 2025; Sheahan & Barrett, 2017; Xue & Liu, 2019). As the global population grows, ensuring a stable food supply and improving storage practices becomes increasingly critical.

Potatoes are a vital crop in both Kazakhstan and the global food supply, ranking as the fourth most-produced food crop worldwide, following rice, wheat, and corn (Birch et al., 2012). They are a

significant source of essential nutrients and a staple food in many countries, playing a pivotal role in addressing food insecurity, particularly in developing regions. Despite their importance, potatoes are highly susceptible to spoilage due to chemical, physical, and microbiological factors. Chemical spoilage, such as the oxidation of phenolic compounds, results in discoloration and texture degradation (Naumann et al., 2020; Sanchez-Reinoso et al., 2020), while physical damage during harvesting and transportation accelerates the production of glycoalkaloids, which can adversely affect potato quality (Okamoto et al., 2020). Moreover, physical damage significantly increases the tubers' susceptibility to microbial infection (Peters, 1996). Microbial spoilage of potatoes, particularly from bacterial and fungal pathogens, remains a major challenge for farmers. Potatoes contain rich stores of starch, polysaccharides, and various proteins, making them an ideal substrate for microbial growth. Facultative anaerobes, such as *Escherichia coli*, other Gram-negative bacteria from the *Enterobacteriaceae* family, and *Coliform* bacteria, as well as fungi like *Aspergillus*, *Cladosporium*, and *Fusarium* spp., are among the most common and destructive agents during storage and post-harvest periods (Blount, 2015a; Doan &

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Davidson, 2000a). These organisms degrade energy-rich nutrients, producing gases such as carbon dioxide and methane, along with various VOCs, including alcohols like ethanol, methanol, and butanol (Ibrahim et al., 2021; D. Li et al., 2021).

Recent research has shifted not only toward the prevention of spoilage (e.g., UV treatment, chemical methods, light exposure) but also toward the early detection of spoilage, particularly using electronic noses (E-noses) (Wilson, 2013; M. Xu et al., 2017). E-noses enable real-time spoilage detection (Fig. S1) by mimicking the human olfactory system through sensor arrays (Seesaard et al., 2022). Various types of gas sensors are integrated into E-noses, including optical sensors (Singh & Gaur, 2023), calorimetric sensors (Wu et al., 2023), surface acoustic wave (SAW) sensors (Lee et al., 1998), and electrochemical sensors (Farquhar et al., 2021; Rutolo et al., 2016). However, metal oxide semiconductor (MOS) based sensors are widely regarded as superior for spoilage detection applications due to their high sensitivity, rapid response times, low cost, and robustness in harsh environments. MOS sensors function by measuring changes in electrical resistance when target gases interact with the sensor surface, typically composed of metal oxides like SnO_2 , (Gautam et al., 2024), ZnO (Soltabayev et al., 2022; Yergaliuly et al., 2025), or WO_3 (X. Li et al., 2024). The gas-sensing mechanism relies on gas adsorption, inducing charge transfer that modulates the depletion layer and sensor conductivity. For instance, MQ-3, MQ-4, MQ-5, MQ-8, MQ-9, and MQ-135 are commonly used sensors for detecting food spoilage in fish (Hasan et al., 2012), meat (Damdam et al., 2023), vegetables (Yavuzer et al., 2024), and fruits (Sonwani et al., 2022), but the MQ-3 has been less studied for real-time monitoring and has not been tested for microbial spoilage in potatoes or compared directly with microbiological methods.

This study aims to develop and validate a low-cost, non-invasive approach for early detection of potato spoilage by integrating a MOS gas-sensor array - centered on an MQ-3 alcohol/VOC transducer - with standard microbiological assays. The work seeks to establish quantitative links between sensor responses and indicator microbial loads, capturing both bacterial and fungal contamination (CFU g⁻¹), and to determine the most suitable sensor configuration and decision criteria for practical, real-time post-harvest surveillance. The study ultimately defines experimental protocols, performance metrics, and operational thresholds to guide deployment in storage and supply-chain settings.

2. Methodology

In order to confirm the gas release during the ripening process, we utilized a gas sensor. Our options were limited to commercially available metal oxide-based gas sensors, specifically the MQ series. For the crea-

tion of the sensor array, we used MQ-3, MQ-4, MQ-5, MQ-8, MQ-9 and MQ-135 sensors (Wisen Sensor MQ type, 2025). The MQ family offers selectivity profiles aligned with spoilage-related volatiles. MQ-3 is most responsive to alcohols (e.g., ethanol) and assorted VOCs; MQ-4 targets methane/natural gas; MQ-5 is tuned to LPG components such as propane and isobutane; MQ-8 preferentially senses hydrogen (of limited relevance to food spoilage); MQ-9 responds to carbon monoxide and other flammables; and MQ-135 provides broad coverage of ammonia, CO_2 , benzene, and multiple VOCs associated with microbial decomposition, as summarized in Table S1. (Farmanesh et al., 2019; Kumar Sai et al., 2019; Wisen Sensor MQ type, 2025). Furthermore, in this study, the Arduino Uno R3 microcontroller board with the ATmega328 microcontroller was used. The Arduino UNO sensor shield was employed to enable simultaneous reading of all connected sensors. As illustrated in Fig. 1, the shield organises each input and output pin into a 3-pin format comprising DATA, VCC, and GND, which is evenly distributed across the board. The schematic connection of the Arduino is illustrated below in Fig. 1. During the sensing process, potatoes are washed tap water and cut into small, uniformly sized pieces (1 cm³). These pieces are then placed inside a sealed box equipped with any other MOS material-based gas sensor. Each day, 2 to 3 pieces of potatoes are removed for microbiological test, this experiment was conducted in two biological replicas. The spoilage of potatoes was assessed through bacterial inoculation and colony enumeration. A 3 g potato sample was homogenized with 15 mL of 0.1 % Buffered Peptone Water (BPW) using Stomacher® 400 Circulator lab blender at approximately 230 rpm for 1 min. Samples were collected at five time points: 0, 1, 3, 5, and 8 days of storage. The homogenized samples were serially diluted to the following dilution factors: 10^{-3} , 10^{-4} , 10^{-6} and 10^{-7} . For microbiological analysis accordingly (Blackburn et al., 1996), 1 mL of each dilution was plated onto *E. coli*/Coliform Count, Total Coliform Count and Aerobic Plate Count petri film (https://www.3m.com, n.d.). These plates were incubated at 37 °C for 24 h. Yeast and mold counts were determined using Petrifilm Yeast and Mold plates, which were incubated at room temperature for 3 days. The formula for colony enumeration:

$$\text{CFU} / \text{g} = \frac{\text{Dilution factor} \times \text{CFU count}}{\text{Sample weight}} \quad [1]$$

The microbiological count results are presented as means and standard errors of means from two independent experiments. Treatment effects were evaluated using ANOVA (JMP 8.0, <http://www.jmp.com>). Comparisons among three or more groups were evaluated using one-way ANOVA followed by Tukey's HSD test.

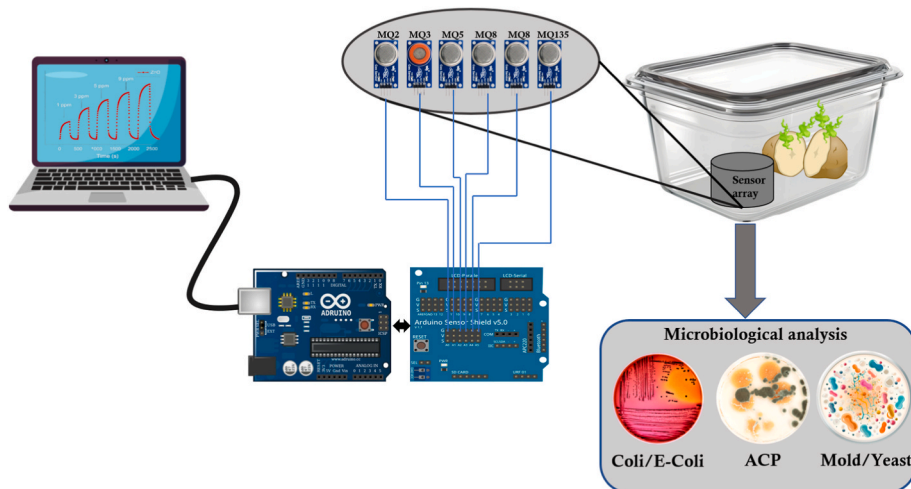


Fig. 1. Schematic illustration of the experimental setup for spoilage detection using Arduino-based gas sensor array and microbiological validation.

3. Results and discussions

3.1. Experimental validation of spoilage gas emission through MQ-3 sensor and microbiological correlation

Early detection of spoilage is critical for reducing food waste and ensuring food safety across supply chains. In this study, to investigate the correlation between gas emissions and microbiological activity during potato spoilage, a controlled experiment was conducted by applying the MQ-3 sensor, which is sensitive to VOCs as depicted in Fig. 2a. The sensor was placed in a sealed container with freshly cut potato samples and connected to an Arduino-based data acquisition system. Microbiological analysis was conducted in parallel to validate the sensor's readings. As illustrated in Fig. 2b, the sensor resistance from Day 0 to Day 3 was around 5.5 kOhm, indicating low levels of alcohol vapour in the box. During the period there is no evidence of spoilage that are observed with sensor data, physical camera and microbiological analysis as shown in Fig. 2c. By Day 3, we observe a steady decrease in resistance, reaching significantly lower values in the final days, which corresponds to an increase in the VOCs in Fig. 2b, particularly ethanol. This rise corresponded with the growth of aerobic bacteria, coliforms, and *E. coli* populations in Fig. 2c. By Day 8, a significant drop in sensor resistance was recorded in Fig. 2b, aligning with the increase of yeast and mold populations in Fig. 2c. Addition, during this period the infrared spectra (Fig. S2) showed no significant changes. These microbial dynamics are typical for raw vegetables like potatoes, which offer a nutrient-rich environment, high water activity, and near-neutral pH, ideal conditions for microbial growth (Doan & Davidson, 2000b). Generally, direct contact with soil and surfaces also contributes to early contamination by bacteria and fungi. In such environments, aerobic bacteria usually grow rapidly within 3–5 days (Blount, 2015b; Dupuis &

Liu, 2019) which aligns with the microbiological increase observed in spoiled potato samples illustrated in Fig. 2d. Notably, this microbial activity and VOC emission occurred before any visible sign, such as discoloration, softening, and fungal growth. The spoilage visual changes of spoilage had only appeared around days 11–12 as depicted Fig. 2d, (yellow arrows). This delay highlights that VOC emission patterns can mirror microbial growth earlier, making VOC monitoring a reliable, non-invasive tool for early spoilage detection.

In general, food spoilage across various products is largely associated with the growth of microorganisms such as *E. coli*, *Enterobacteriaceae*, coliforms and molds (Luna-Guevara et al., 2019). These microorganisms are widely used as indicators of food spoilage and serve as important markers for assessing food quality and hygiene during production and handling (Shah et al., 2023). Indicator organisms can also help predict the potential presence of harmful pathogens in the food prior to conducting expensive and time-consuming microbiological analyses. Here some common foodborne pathogens include *Staphylococcus aureus*, *Pseudomonas* spp., *Clostridium* spp., *Salmonella* spp., *Listeria monocytogenes*, *Erwinia carotovora*, *E. coli* O157:H7, and *Candida albicans* (Snyder & Worobo, 2018). Notably, *Staphylococcus aureus*, *Pseudomonas* spp., *E. coli* O157:H7 and other bacteria are known to produce VOCs such as indole, methyl ketones, and sulfur-containing compounds during food spoilage (F. Zhang et al., 2025). Meanwhile, *Candida albicans* emit ethanol, ethyl acetate, and aromatic compounds during the food products' decomposition (Coelho Júnior et al., 2019). Recent targeted biosensing advances demonstrate ultrasensitive detection of specific hazards, including UCNP-based platforms for *S. aureus* in meat (Ouyang et al., 2023), zein-film/UCNP solid-phase sensors for nitrosamine contaminants (Ouyang et al., 2024), and RPA-CRISPR/Cas12a assays for *S. aureus* (Guo et al., 2025) - as synthesized in reviews of nanotechnology-driven enzyme/electrochemical and nano-optical

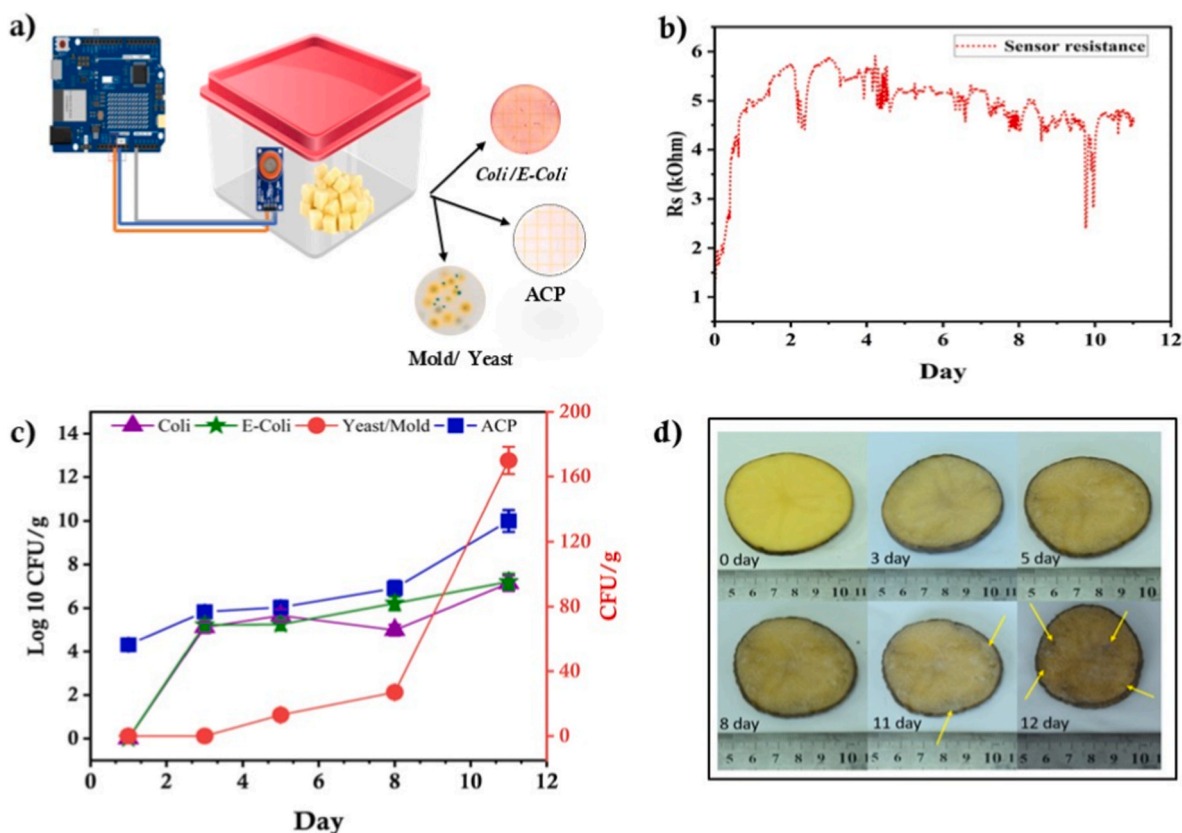


Fig. 2. a) Schematic connection of Arduino UNO and MQ-3 sensor; b) Sensor resistance changes during the spoilage process of stored potatoes; c) Microbiological analysis showing the presence and growth trends of coliform bacteria (*Coli*), *Escherichia coli* (*E. coli*), aerobic colony-forming units (ACP), and Mold/Yeast over time; d) Time-lapse images showing the visual appearance of cut potato samples from day 0 to day 12 in the storage box.

strategies (Qu et al., 2025; Y. Xu et al., 2023). Investigating the sensitivity of VOC sensors and other detection technologies to the specific emissions of harmful foodborne pathogens could provide valuable insights for real-time monitoring in storage environments. Additionally, it is important to evaluate whether these sensors can reliably differentiate between various pathogens as accurately as conventional microbiological methods.

3.2. Application of MQ-3 sensor in different storage conditions

To assess the impact of environmental conditions on spoilage, three treatments were tested over a 15-day period: untreated (control), high humidity (~80 %), and UV irradiation (30-min pretreatment). Spoilage progression was evaluated using MQ-3 sensor resistance and microbiological analyses. The time-dependent resistance measurements from the sensor showed clear spoilage patterns among the groups in Fig. 3a. The MQ-3 gas sensor showed distinct response patterns among the three groups (Fig. 4a). In both the control and high-humidity conditions, resistance dropped rapidly in the early days, suggesting a sharp increase in VOCs (Fig. 4a, black and red lines), such as ethanol, likely produced by microbial growth. In contrast, UV-treated samples maintained higher resistance values throughout the experiment, illustrated in Fig. 3a (blue line), indicating a reduced release of spoilage-related VOCs. This indicates that UV treatment successfully inhibited early microbial growth, thereby postponing the generation of gases associated with spoilage. Microbial counts obtained on Day 1 and Day 15 confirmed the sensor findings (Figs 4b–d). In the aerobic plate count (APC), *E. coli*/coliform counts and mold/yeast counts a significant increase was observed in the control and high-humidity group demonstrated in Fig. 3b–d, respectively. Moisture facilitated spore germination and hyphal expansion, consistent with known optimal humidity ranges for fungal development

(85–95 %) (Hammond et al., 2015). Additionally, high humidity accelerated microbial growth, due to promotion of bacterial motility, biofilm formation, and fungal colonization (Y. Zhang et al., 2023). In contrast, UV pretreatment significantly suppressed microbial activity (Fig. 3b–d). These results show that the VOC sensor responses closely reflected the microbiological trends under the different treatments application in storage box, supporting the potential of the MQ-3 sensor for non-invasive spoilage monitoring.

3.3. Non-invasive monitoring of potato degradation using a multi-gas sensor array

To assess the performance of MQ-series gas sensors in a potato-spoilage model, we conducted a comparative experiment under two conditions: (i) ambient air as the sensing baseline and (ii) a sealed box containing cut potatoes to induce and monitor spoilage. The resistance values of six sensors - MQ-3, MQ4, MQ5, MQ8, MQ9, and MQ-135 were continuously monitored over several days illustrated Fig. 4a and b. During exposure to ambient air and relative humidity (RH) around 30 %, all sensors displayed relatively stable resistance profiles, indicating minimal interference from environmental gases. This baseline behavior confirms their reliability and stability under uncontaminated conditions. Meanwhile, under the spoilage conditions, an alteration in sensor resistance was observed, allowing for the detection of potato spoilage. A gradual decrease in the resistance of sensors (MQ-3, MQ-9, and MQ-135) starting from the second day indicates exposure to VOCs, NH_3 , and CO_2 gases, which are commonly emitted during the microbial degradation of food (Binson & Thomas, 2023; John et al., 2021). The early response of MQ-3 and MQ-135, in particular, suggests strong sensitivity to spoilage-related gases, with MQ-3 showing a sharper resistance drop and MQ-135 exhibiting a more gradual but sustained decrease. A little later,

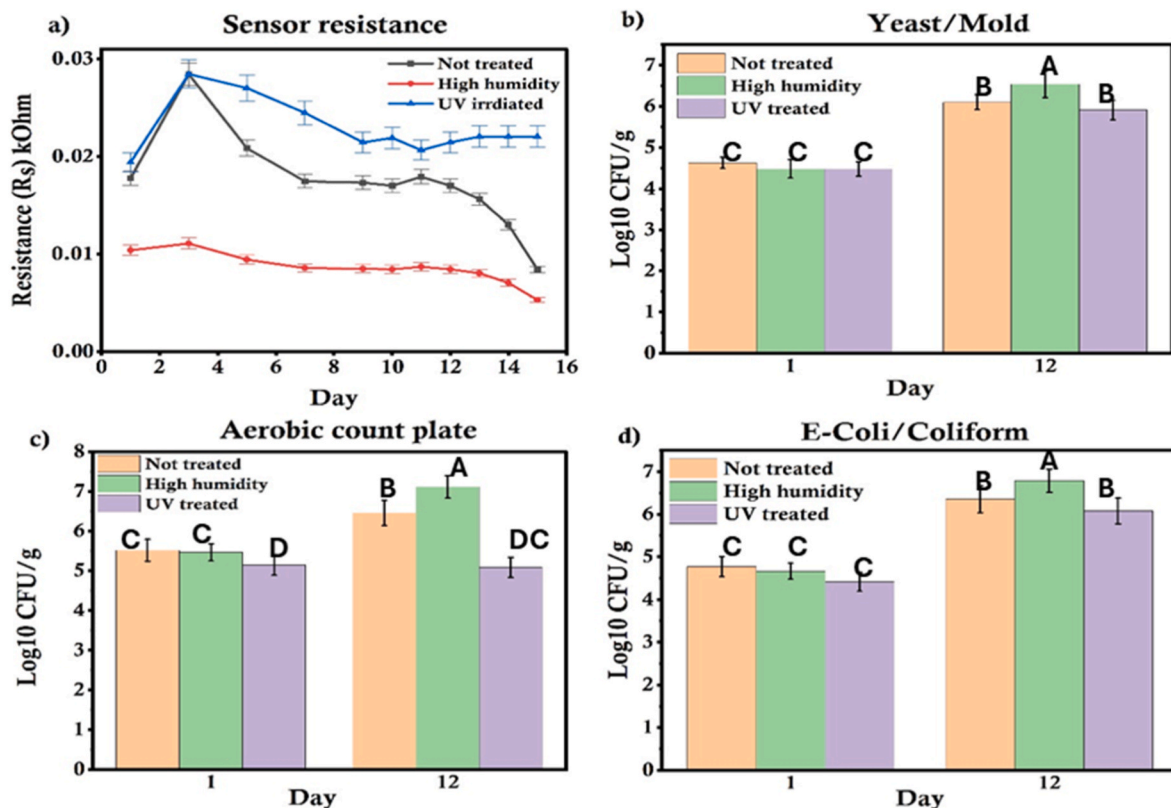


Fig. 3. a) Time-dependent changes in sensor resistance over 15 days for three storage groups (Not Treated, High humidity (~80 %), UV-treated (30 min)). b–d) Microbiological analysis at the end of the storage period showing total coliforms (*Coli*), *E. coli*, aerobic mesophilic bacteria (APC), and fungi levels. Error bars indicate standard deviation. Each group is color-coded: control (orange), group 1 (green), and group 2 (purple). Different letters indicate significant differences between means as determined using ANOVA followed by a Tukey's HSD test ($P < 0.05$).

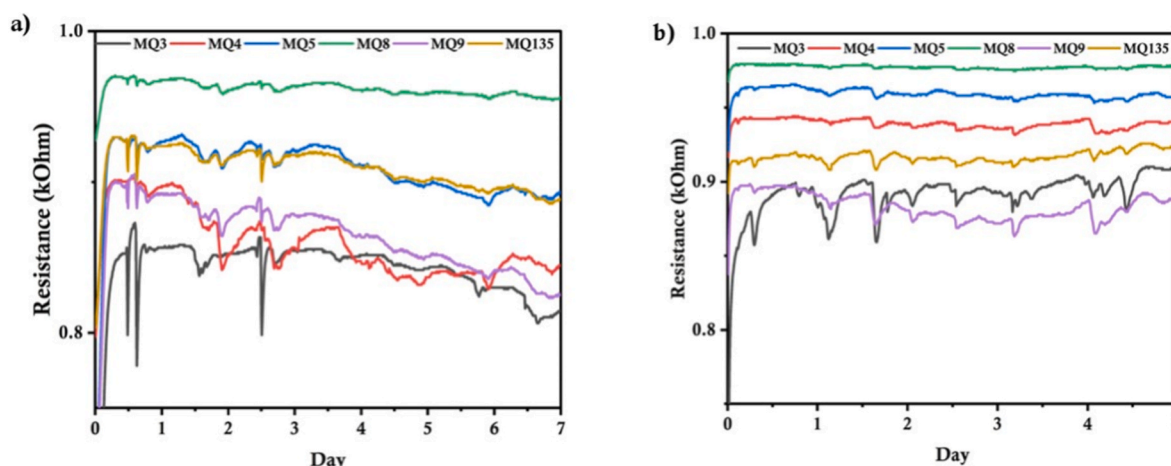


Fig. 4. MQ sensors monitoring of the spoilage of the potatoes: a) baseline air monitoring, b) detection during spoilage.

between days 2.5 and 3, the readings from the MQ-4 and MQ-5 sensors also started to decline slightly. This slower response is likely due to their sensitivity to hydrocarbon gases. The MQ-8 gas sensor showed minimal response to the spoilage of potatoes, as it is selective for hydrogen gas, which is not a primary gas produced during organic degradation. Based on these observations, the spoilage detection threshold was defined as the point at which sensor resistance continuously decreased for two or more consecutive days. Among the sensors tested, MQ-3 and MQ-135 proved to be the most responsive. The MQ-3 sensor exhibited a sharp drop in resistance beginning on day 2, while MQ-135 showed a gradual but consistent decline. These findings align with the study conducted by Yavuzer (2021), which demonstrated similar sensitivity patterns in MQ sensors used only for fish spoilage detection (Yavuzer, 2021) and without detailed microbiological test. In their research, analog voltage readings from the MQ-3 and MQ-135 sensors increased significantly at the onset of spoilage, correlating with microbial counts exceeding 7 log CFU/g (Yavuzer, 2021). Although the microbial spoilage mechanisms differ between starch-based vegetables like potatoes and protein-rich fish, our findings reinforce that MQ-3 and MQ-135 sensors are consistently reliable for detecting early spoilage across these distinct food types. In our work, we further refined the spoilage detection criteria by defining it as a sustained resistance decline for two or more consecutive days, with MQ-3 and MQ-135 providing the most responsive and clear indicators of deterioration.

Overall, the comparative analysis confirms that MQ-3 and MQ-135 sensors are highly effective for early detection of potato spoilage, demonstrating reliable sensitivity to microbial degradation gases, and thereby validating their suitability for low-cost, real-time food quality monitoring applications. Meanwhile, other MQ sensors are not applicable for real-time spoilage detection. In the future, we plan to develop our own sensing material based on MOS with enhanced sensitivity and stability to further improve the performance of real-time potato spoilage detection systems.

4. Conclusion

This study shows that the MQ-3 sensor, part of a commercial MOS gas sensor array, is effective for real-time, non-invasive detection of potato spoilage under various storage conditions, correlating strongly with microbiological analyses and identifying spoilage before visible degradation occurs. Comparative tests across ambient, high-humidity, and UV-treated conditions further confirmed the sensor's ability to monitor and differentiate spoilage progression based on VOC emission patterns. Among the six sensors tested, the MQ-3 sensor consistently displayed the most sensitive and timely response, performing on par with MQ-135. In contrast, the other sensors MQ-4, MQ-5, MQ-8, and

MQ-9 did not demonstrate an early response like the MQ-3 for potato spoilage. This makes the MQ-3 an affordable and practical candidate for early spoilage detection in potatoes. These findings support the integration of gas sensor arrays as a complementary or alternative tool to traditional microbiological methods for enhancing post-harvest monitoring, minimizing food waste, and improving storage management. Future research will focus on developing enhanced MOS-based sensing materials to further improve selectivity, sensitivity, and long-term stability for broader food quality monitoring applications.

CRediT authorship contribution statement

A. Tangirbergen: Writing – review & editing, Writing – original draft, Software, Investigation. **A. Tleubekova:** Writing – review & editing, Writing – original draft, Methodology, Data curation. **G. Yergaliuly:** Writing – review & editing, Methodology, Investigation. **A. Kurmanbayeva:** Writing – review & editing, Investigation, Formal analysis. **B. Soltabayev:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **A. Soltabayeva:** Writing – review & editing, Visualization, Validation, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.lwt.2025.118926>.

Data availability

Data will be made available on request.

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